Name:Vandhana M Register Number: 250701804 Experiment Name: Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# ------------------------------------------------
# 1. CREATE DATASET
#-------------------------------------------------
data = {
    "Date":[
        "01-01-2023", "02-01-2023", "03-01-2023", "04-01-2023",
        "05-01-2023", "06-01-2023", "07-01-2023", "08-01-2023",
        "09-01-2023", "10-01-2023", "11-01-2023", "12-01-2023",
        "13-01-2023", "14-01-2023", "15-01-2023", "16-01-2023"
    ],
    "Product": [
        "Product A", "Product B", "Product A", "Product C",
        "Product B", "Product A", "Product C", "Product B",
        "Product A", "Product C", "Product B", "Product A",
        "Product C", "Product B", "Product A", "Product C"
    ],
    "Sales": [
        200, 150, 220, 300, 180, 210, 320, 160,
        230, 310, 190, 240, 330, 170, 250, 340
    ],
    "Quantity": [
        4, 3, 5, 6, 4, 5, 7, 3,
        6, 7, 4, 6, 8, 3, 7, 8
    ],
    "Region": [
        "North", "South", "North", "East",
        "West", "North", "East", "South",
        "North", "East", "West", "North",
        "East", "South", "North", "East"
    ]
}
# Create DataFrame
df = pd.DataFrame(data)
# Convert Date column
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)
# ------------------------------------------------
# 2. DISPLAY DATA
# ------------------------------------------------
print("DATASET:")
print(df.head())
print("\n")
# ------------------------------------------------
# 3. CHECK MISSING VALUES
```

```
# ------------------------------------------------
print("MISSING VALUES:")
print(df.isnull().sum())
print("\n")
# ------------------------------------------------
# 4. STATISTICAL SUMMARY
# ------------------------------------------------
print("STATISTICAL SUMMARY:")
print(df[["Sales", "Quantity"]].describe())
print("\n")
# ------------------------------------------------
# 5. PRODUCT-WISE SALES AND QUANTITY
# ------------------------------------------------
product_summary = df.groupby("Product")[["Sales",
"Quantity"]].sum().reset_index()
print("PRODUCT-WISE SALES AND QUANTITY:")
print(product_summary)
print("\n")
# ------------------------------------------------
# 6. LINE GRAPH - SALES OVER DATE
# ------------------------------------------------
plt.figure(figsize=(10, 5))
plt.plot(
    df["Date"],
    df["Sales"],
    marker="o",
    color="blue",
    linewidth=2
)
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
# ------------------------------------------------
# 7. BAR GRAPH - PRODUCT-WISE SALES
# ------------------------------------------------
plt.figure(figsize=(8, 5))
plt.bar(
    product_summary["Product"],
    product_summary["Sales"],
    color=["blue", "orange", "green"]
)
plt.title("Product-wise Total Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.tight_layout()
```

```python
plt.show()
# ------------------------------------------------
# 8. CORRELATION MATRIX
# ------------------------------------------------
correlation = df[["Sales", "Quantity"]].corr()
print("CORRELATION MATRIX:")
print(correlation)
print("\n")
# ------------------------------------------------
# 9. CORRELATION HEATMAP
# ------------------------------------------------
plt.figure(figsize=(7, 5))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    linewidths=0.5,
    fmt=".2f"
)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

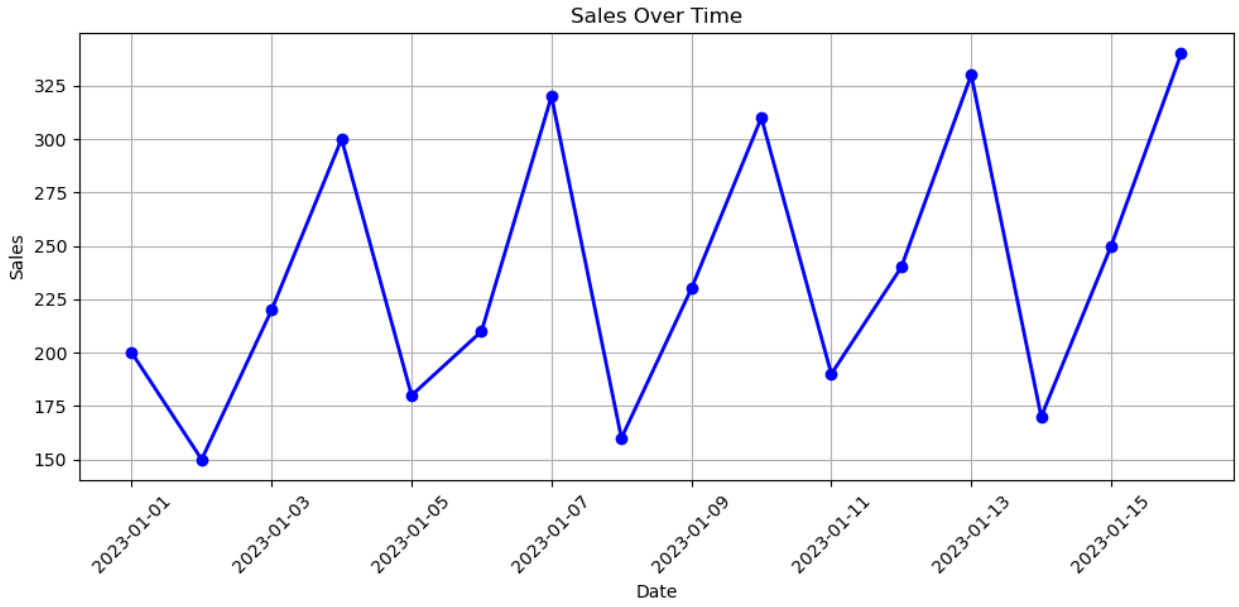
```
DATASET:
        Date    Product  Sales  Quantity Region
0 2023-01-01  Product A    200         4  North
1 2023-01-02  Product B    150         3  South
2 2023-01-03  Product A    220         5  North
3 2023-01-04  Product C    300         6   East
4 2023-01-05  Product B    180         4   West

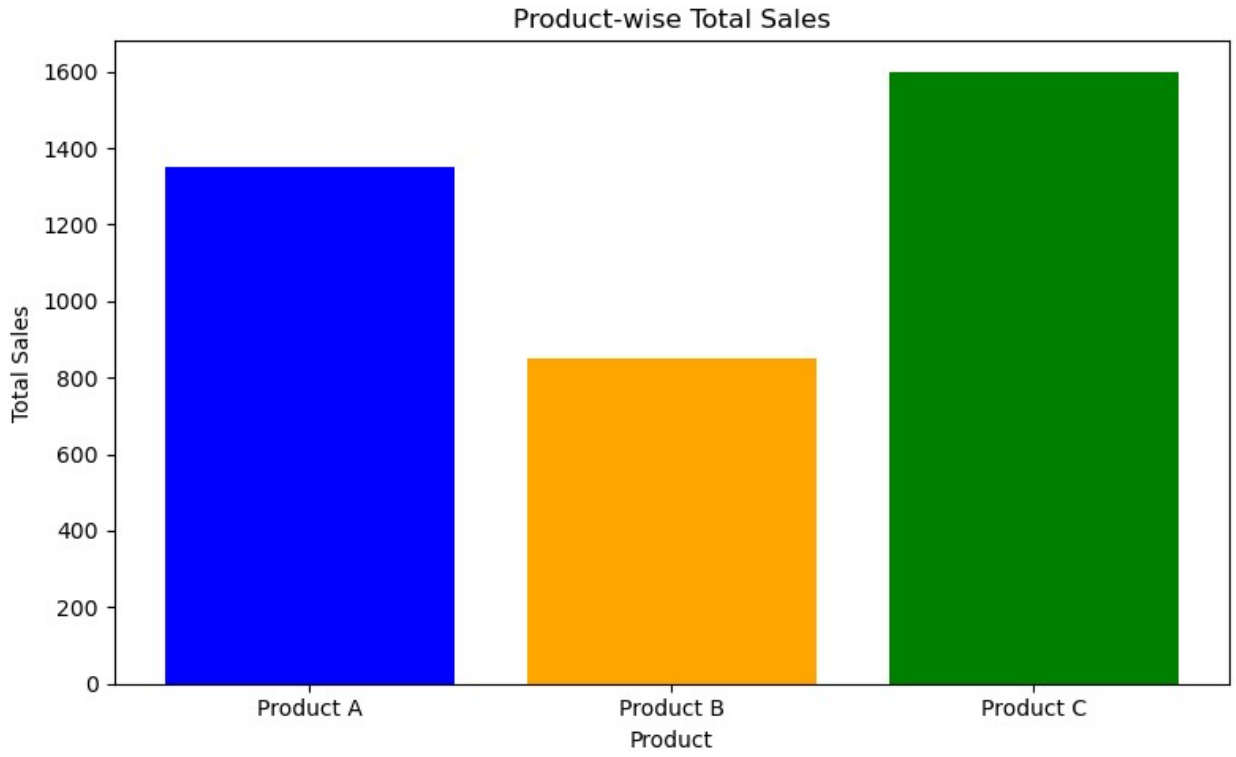

MISSING VALUES:
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64


STATISTICAL SUMMARY:
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
```

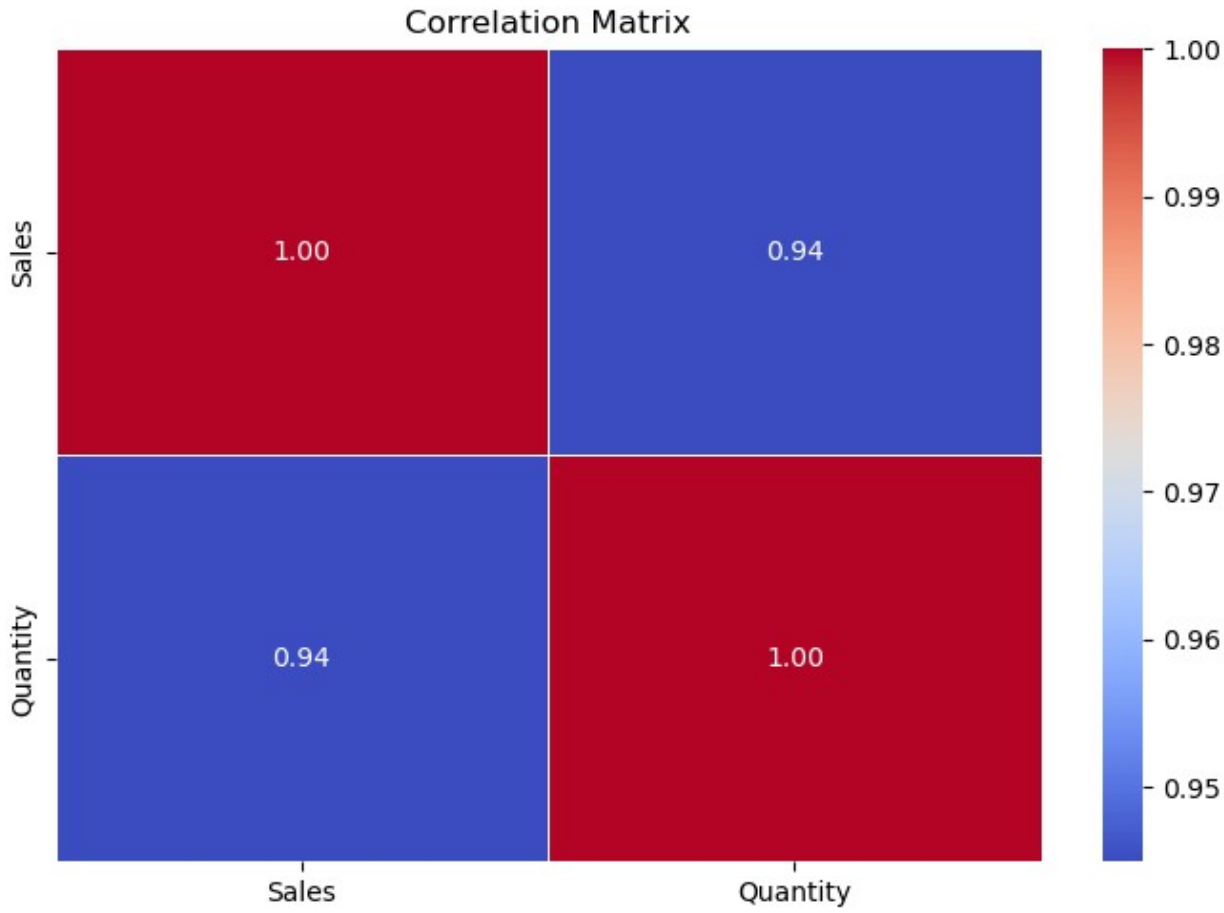
```
max    340.000000   8.000000


PRODUCT-WISE SALES AND QUANTITY:
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
CORRELATION MATRIX:
              Sales  Quantity
Sales     1.000000  0.944922
Quantity  0.944922  1.000000
```


Correlation Matrix
Sales
Quantity
1.00
0.94
0.94
1.00
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95